INCH-POUND

GGG-P-453B
18 November 2025
SUPERSEDING
GGG-P-453A
8 April 1965

FEDERAL SPECIFICATION

PLATE, SURFACE, CAST IRON

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast iron surface and bench plates for general inspection and layout work (see 6.1).

1.2 Classification.

1.2.1 Types and grades. Surface plates covered by this specification may be of the following types and grades, as specified (see 6.2):

Type I – Single surface.
Grade A – Fine scraped.
Grade B – Scraped.
Grade C – Fine planed.
Grade E – Ground.
Type II – Double surface.
Grade A – Fine scraped.
Grade B – Scraped.
Grade E – Ground.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services

(Copies of this document are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5098 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5220

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E10 - Standard Test Method for Brinell Hardness of Metallic Materials

(Copies of this document are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, W. Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY OF MECHANICAL ENGEERS (ASME)

ASME-B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available from www.asme.org or ASME International, Three Park Avenue, New York, NY 10016-5990.)

MASTER PAINTERS INSTITUTE (MPI)

MPI #9 - Alkyd, Exterior Gloss (MPI Gloss Level 6)
MPI #94 - Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)

(Copies of these documents are available from https://mpi.net.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive, and are not intended to preclude the purchase of surface plates otherwise conforming to this specification.

3.1.1 Availability. Inspection and layout plates required in accordance with this specification shall be products normally manufactured by the prospective producer. These items produced shall also be available on a commercial basis.

3.2 Construction. Construction shall be free from any characteristics or defects that may render the surface plate unsuitable or inefficient for the purpose intended.

3.3 Material. The plates covered by this specification shall be made of fine grain gray iron or iron alloy castings.

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3.3.1 Reclaimed materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

3.3.1.1 Seasoning. All castings used in the fabrication of the surface plates covered by this specification after rough machining shall be stress relieved before final finishing. Stress relieving procedures shall be of the method normally used by the producer which will produce surface plates of the accuracy specified herein.

3.4 Hardness. The finished surface plates shall have a Brinell hardness of not less than 180 when tested as specified in 4.4.1.

3.5 Working surface. The working surface of all surface plates covered by this specification shall extend beyond the ribbing a minimum of one inch, exclusive of fillets, to permit the use of clamps. The working surface shall be of the following grades as specified (see 6.2).

3.5.1 Grade A. Grade A working surfaces shall be fine precision scraped with relief spots to prevent the sticking of precision instruments and gage blocks. The spotting cavities shall be from 0.0002 to 0.0003 inches in depth with a concentration of 15 to 18 spotting cavities per square inch and a 20 to 40 percent bearing area surface of texture requirements of table I.

3.5.2 Grade B. Grade B working surface shall be scraped with relief spots to prevent the sticking of precision instruments and gage blocks. The spotting cavities shall be from 0.0002 to 0.0004 inches in depth with a concentration of 8 to 11 spotting cavities per square inch and 10 to 40 percent bearing area surface of texture requirements of table I.

3.5.3 Grade C. Grade C working surface shall be smooth machined to texture requirements of table I. Tool marks shall not exceed 0.0005 inches in depth.

3.5.4 Grade E. Grade E working surface shall be precision ground to a surface finish of 32 microinches arithmetic accuracy (AA) or better in conformance with ASME-B46.1. Spotting cavities shall be from 0.0002 to 0.0003 inches in depth with a concentration of 15 to 18 spotting cavities per square inch and a 20 to 40 percent bearing area.

3.6 Accuracy. The bearing areas of the working surfaces shall not deviate from a mean plane by more than that specified in table I when tested as specified in 4.4.2. No adjacent square foot areas shall vary with each other by more than the following:

Grades A and E	0.0001 inch.
Grade B	0.0002 inch.
Grade C	0.0003 inch.

3.6.1 Side edges. The side edges of the top shall be parallel to each other and square with the end edges Within 0.0002 inches per linear foot and all edges shall be square with the working surface within 0.0002 inch in the full thickness of the top.

3.7 Handling provisions. Each plate shall be provided with lifting handles or other devices in accordance with the size of plate for lifting and transporting. The lifting device shall be so located as to reasonably balance the plate when suspended.

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Table I. Requirements for bearing areas.

	Maximum plus or minus deviation from mean plane and roughness* height ratings as per ASME-B46.1 (Microinches)			Maximal allowable plus or minus deviation tolerances for					
				Repeat reading measurements (Microinches)			Total flatness of entire plate (Microinches)		
Diagonal Range (Inches)	GRADE A	GRADE B	GRADE C	GRADE A & E	GRADE B	GRADE C	GRADE A & E	GRADE B	GRADE C
Through 25	50	100	200	50	100	200	100	150	300
Over 25 Through 40	75	150	300	75	150	300	150	225	450
Over 40 Through 50	100	200	400	100	200	400	200	300	600
Over 50 Through 60	150	300	600	150	300	600	300	450	900
Over 60 Through 70	200	400	800	200	400	800	400	600	1200
Over 70 Through 85	250	500	1000	250	500	1000	500	750	1500
Over 85 Through 100	350	700	1400	350	700	1400	700	1050	2100
Over 100 Through 125	500	1000	2000	500	1000	2000	1000	1500	3000
Over 125 Through 150	750	1500	3000	750	1500	3000	1500	2250	4500
Over 150	1100	1500	3000	1100	1500	3000	2200	2250	4500

*Above values apply to maximum roughness.

3.8 Screw threads. All screw threads used on surface plates shall be in accordance with the screw threads covered in FED-STD-H28.

3.9 Finish. External surfaces shall be smooth and clean with all edges rounded or beveled. The side and end edges of the working surfaces and the bearing pads shall be smooth machine finished to 125 microinch AA or better in accordance with ASME-B46.1.

3.10 Painting. Unless otherwise specified (see 6.2), all surfaces not machined shall be painted in accordance with manufacturer's normal practice.

3.11 Ribbing. Ribbing, of sufficient thickness, shall be incorporated into the design of surface plates to assure maximum flatness, squareness, and stability of all working surfaces.

3.12 Type I, single surface plates. Type I surface plates shall be a single casting with diagonal cross ribbing providing three point bearing support at the junction of the ribbing. Type I surface plates shall have a single working surface. Accuracy of the working surface shall conform to the requirements specified herein for the grad specified (see 6.2). Type I surface plates shall be similar to figure 1, and shall conform to the dimensions of table II for the size specified (see 6.2).

3.12.1 Covers. All type I, grades A, B, and E surface plates shall be furnished with a properly fitted wooden cover so designed as to protect the edges as well as the working surface. Each cover shall be given at least two coats of varnish or shellac.

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3.13 Type II, double surface plates. Type II double surface plates shall have working surfaces on the top and bottom. Surface plates of this type shall be approximately 10 inches wide by 14 inches long by 2.375 inches in depth, weigh approximately 40 pounds, and be in substantial agreement with figure 2. The working surface of type II double surface plates shall be grade A, B, or E as specified (see 6.2).

3.14 Nameplate. Unless otherwise specified (see 6.2), a metal nameplate permanently and legibly marked with the following information shall be securely attached to each surface plate in a prominent location:

Fed. Spec. GGG-P-453
Type
Grade
Size
Manufacturer's serial number
(Manufacturer's name or trademark)

3.15 Workmanship. Workmanship of the surface plates shall be of a quality prevailing among manufacturers normally producing surface plates of the type and sizes specified herein.

Table II. Type I Surface plate, sizes and dimensions.

Nominal Size		Minimum Top Solid Thickness (Inches)	Minimum Height (Inches)	Approximate Diagonals (Inches)
Width (Inches)	Length (Inches)			
8	12	0.500	2.500	14.4
10	14	0.500	2.500	17.2
12	14	0.500	3	18.4
12	18	0.625	3.250	21.6
12	24	0.625	3.250	26.8
14	18	0.625	3.500	22.8
16	22	0.625	3.500	27.2
18	18	0.625	3.500	25.5
18	24	0.688	4.250	30
18	30	0.688	4.250	34.9
18	36	0.688	4.750	40.3
24	24	0.688	4.750	33.9
24	30	0.750	4.750	38.4
24	36	0.750	6	43.3
24	48	0.750	6.500	53.7
30	48	1	6.750	56.6
30	60	1	6.750	67.1
36	36	1	7	50.9
36	48	1	7	60
36	72	1	9	80.5
48	96	1.125	10	107.3

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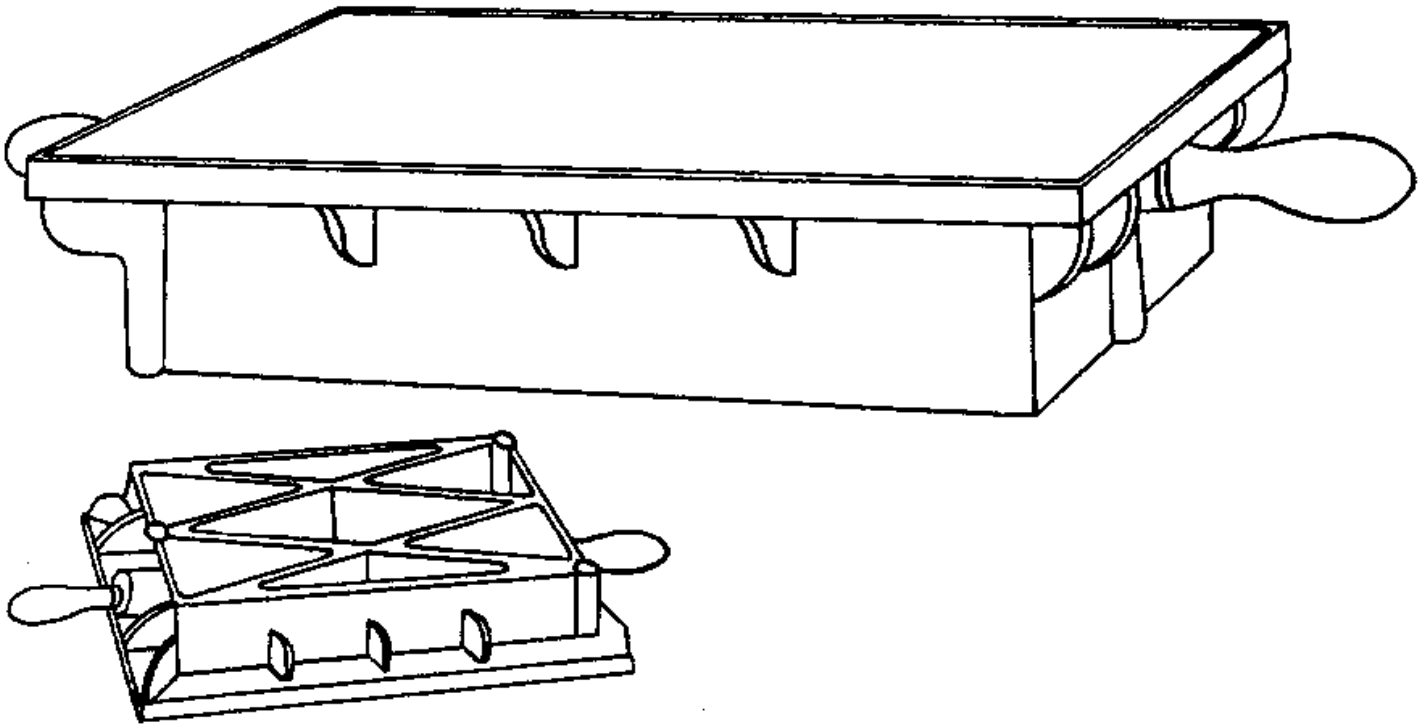


Figure 1. Type I surface plate.

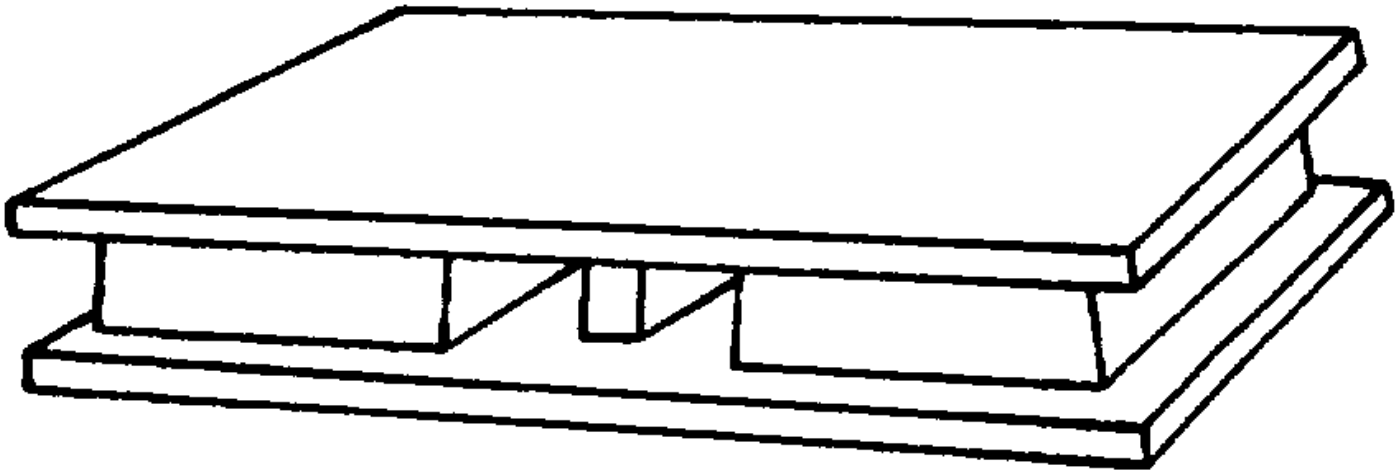


Figure 2. Type II surface plate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All surface plates of the same type, grade, and size presented at the same time shall be considered a lot for the purposes of inspection.

4.2.2 Sampling for examination and tests. A random sample of surface plates shall be selected for each lot offered for quality conformance inspection in accordance with ASQ/ASME-Z1.4, inspection level II.

4.3 Quality conformance inspection. Each of the sample surface plates selected in accordance with 4.2.2 shall be inspected to verify compliance with this specification. Inspection for quality conformance shall be as follows:

(a) Examination (see 4.3.1).
(b) Tests (see 4.3.2 and 4.4).

4.3.1 Examination. Each of the surface plates selected in accordance with 4.2.2 shall be examined for conformance with the requirements of this specification. Examination shall be conducted as specified in table III. The acceptable quality levels (AQL) shall be as indicated in table III. Any surface plate containing one or more defects shall not be offered for delivery and if the number of nonconforming plates exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 Tests. Each of the sample surface plates selected in accordance with 4.2.2 shall be tested in accordance with 4.4 to determine compliance with the requirements of this specification. The AQL shall be 1.5 percent defective. Any plate failing the test shall not be offered for delivery and if the nonconforming plates in any one sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 Hardness test. Hardness of the surface plate shall be determined in conformance with ASTM-E10 using a 3,000 kg. load and a 10 mm. diameter ball. Readings shall be taken on sections adjacent to the working surface. Preparation of test surfaces shall be made in accordance with ASTM-E10. Failure of the plate to meet the minimum requirements of 180 Brinell hardness number shall be cause for rejection.

4.4.2 Accuracy test of working surfaces. Accuracy of the working surfaces shall be determined by checking flatness of the working surface for conformance with the tolerances specified herein (see 3.6). Methods used in determining conformance shall be an approved method, standard with industry and acceptable to the procuring agency. Failure to meet the accuracy requirements specified shall be cause for rejection.

4.4.3 Side edge parallelism and finish. Parallelism of side edges and squareness of the working surface with side edges shall be determined to comply with the requirements of 3.6.1. Finish of the side edges shall be tested to determine compliance with the 125 microinch AA as specified in 3.9. Failure to meet these requirements shall be cause for rejection.

4.4.4 Bearing areas. The percentage of bearing area shall be determined by very lightly coating the working surface with Prussian blue or other suitable material, then rubbing over the surface with another plate of known accuracy to bring out the high spots. A plate of glass or other suitable flat transparent material that has previously been ruled into small predetermined squares shall then be laid on the working surface and the percent of bearing determined as well as the number of spots per square inch. Bearing areas of the working surfaces shall not exceed the requirements specified in table I for the grade and size tested. Failure to meet these requirements shall be cause for rejection.

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Table III. Classification of defects in accordance with ASQ/ASME-Z1.4.

Categories	Defects
Critical:	None defined.
Major:	AQL – 1.5 percent defective.
101	Surface plates not the grade and type specified.
102	Construction of surface plates not free from any characteristic or defect that might render the plate unsuitable or inefficient for the purpose intended.
103	External surfaces of plates not smooth and clean, all edges not rounded or beveled; the side and end edges of the working surfaces and the bearing pads not smooth machine finished.
104	Casting not free from harmful blowholes, porosity, hard spots, shrinkage or other injurious defects; castings not thoroughly cleaned of all scale, molding sand and other extraneous materials; evidence of flaws in castings repaired by plugging, peening or welding except as authorized.
105	The working surfaces of all surface plates extend beyond the ribbing less than the specified minimum.
106	Ribbing not incorporated into the design of the surface plates as specified; ribbing not of sufficient thickness to assure maximum flatness, squareness and stability of all working surfaces.
107	Type I single surface plates nonconforming, not with diagonal cross ribbing; dimensions of plates not as specified or less than the specified minimum (as applicable); surface plates not designed with three point bearing supports; bearing supports not located as specified.
108	Type II double surface plates nonconforming, dimensions not as specified; working surfaces not as specified.
109	Each surface plate not provided with lifting handles or other lifting devices as specified; lifting devices not located so as to assure reasonable balance of the plate when lifted.
Minor:	AQL – 2.5 percent defective.
201	Fitted wooden covers not furnished with all type I, grades A, B, and E surface plates as specified; wooden covers not painted as specified.
202	All unmachined surfaces of plates not painted as specified.
203	Marking of manufacturer's name or trademark and serial number, and type, grade and size of surface plate, missing, nonconforming, illegible or not permanent; nameplate not securely attached to each surface plate as specified.
204	Evidence of flaws or blemishes of the finished surface plate that might affect the serviceability or appearance.
205	Threaded parts nonconforming, form and class of fit not specified; threads missing, stripped, crossed, torn, or damaged.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. The surface plates covered by this specification are for use in inspection and layout work and to provide a true and rigid surface for assembly work.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and grade (see 1.2, 3.5, 3.12 and 3.13).
c. Size of working surface (see 3.12 and 3.13).
d. Painting, if different (see 3.10).
e. Nameplate requirements, if different (see 3.14).
f. Packaging requirements (see 5.1).

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Plates, layout surface or surface, steel. Carload minimum weight 30,000 pounds.
Motor:
Plates, layout surface or surface steel. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 Subject term (key word) listing.

Double surface
Fine planed
Fine scraped
Ground
Planed
Scraped
Single surface

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Table IV. Metric conversion factors.

Microinches	Microns (Metric)	Inches	MM (Metric)	Inches	Meters (Metric)
50	1	0.500	12.7	40	1.016
75	2	0.625	15.875	40.3	1.02362
100	3	0.688	17.4625	43.3	1.09982
150	4	0.750	19.05	48	1.2192
200	5	1	25.4	50	1.27
225	6	1.500	38.1	50.9	1.29286
250	6	2.375	60.325	53.7	1.36398
300	8	2.500	63.5	56.6	1.43764

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Table IV. Metric conversion factors.(continued)

Microinches	Microns (Metric)	Inches	MM (Metric)	Inches	Meters (Metric)
350	9	3	76.2	60	1.524
400	10	3.250	82.55	67.1	1.70434
450	11	3.500	88.9	70	1.778
500	13	4.250	107.95	72	1.8288
600	15	4.750	120.65	80.5	2.0447
700	18	6	152.4	85	2.159
750	19	6.500	165.1	96	2.4384
800	20	6.750	171.45	100	2.54
900	23	7	177.8	107.3	2.72542
1000	25	8	203.2	125	3.175
1050	27	9	228.6	150	3.81
1100	28	10	254		
1200	30	12	304.8		
1400	36	14	355.6		
1500	38	14.4	365.76		
2000	51	16	406.4		
2100	53	17.2	436.8		
2200	56	18	457.2		
2250	57	18.4	467.36		
3000	76	21.6	548.64		
3300	84	22	558.8		
4400	112	22.8	579.12		
4500	114	24	609.6		
6600	168	25	635		
		25.5	647.7		
		26.8	680.72		
		27.2	690.88		
		30	762		
		33.9	861.06		
		34.9	886.46		
		36.0	914.4		
		38.4	975.36		

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